

NASK NLP Seminar

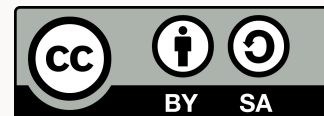
17 Aug 2026

Reasoning Gets Harder for LLMs Inside A Dialogue

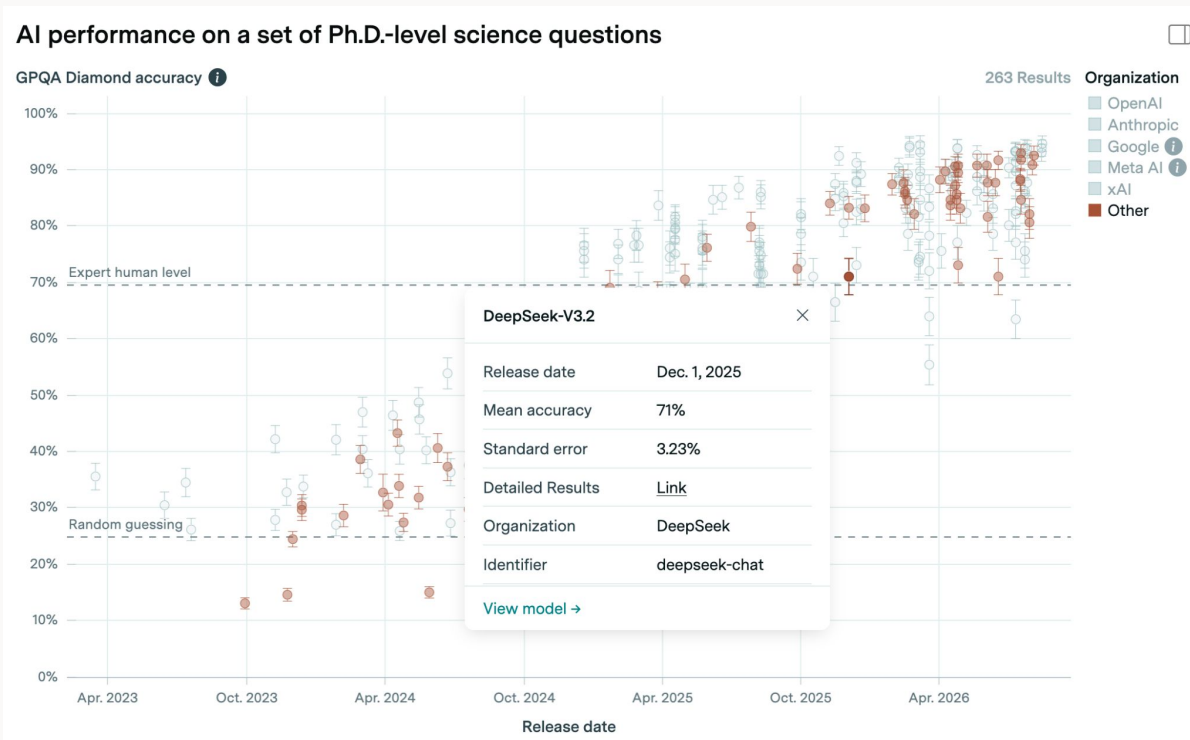
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Faculty of Mathematics and Physics
Institute of Formal and Applied Linguistics



Reasoning benchmarks



Reasoning benchmarks

- Well-defined isolated tasks
- Typically multiple-choice questions
- Static (leading to data contamination)

How does **reasoning** performance on such benchmarks transfer to reasoning **embedded in natural language generation**?

We explore this through:

- (1) Embedding reasoning problems in a task-oriented dialogue (TOD) setup
- (2) Controlled comparison of the standalone and TOD setups

(a) Standalone



Given the list of trains in JSON format below, select departure time of the latest train that arrives in Cambridge before sunset. Current date is 2025-08-27 and current time is 06:05. Sunset time is 19:57.
Trains: [{ "id": "TR5972", "departure": ... }]

(b) Embedded in TOD



You are a helpful assistant specialized in providing travel guidance for Cambridge. [...]
Current date is 2025-08-27 and current time is 06:05.
Sunset time is 19:57.



I need a train departing Cambridge and arriving at London Liverpool Street today. The earliest I can leave is 5:10pm.

[{"function": {"name": "search_trains", ...}}]



[{"train_id": "TR5972", "departure": ...}]

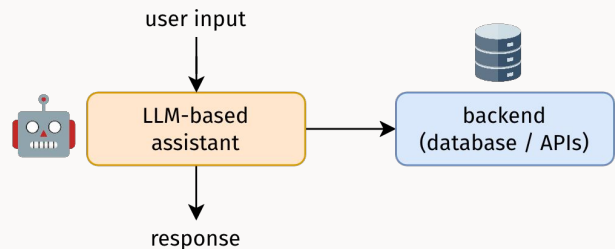
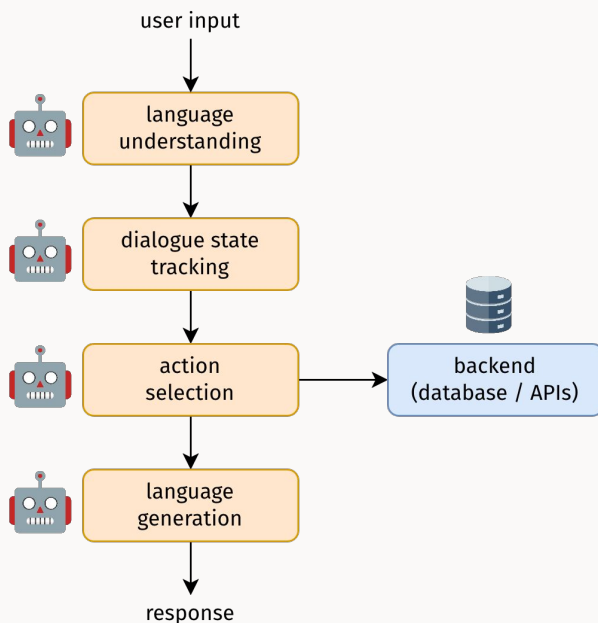
I found four trains departing today starting from 5:10pm.



What time is the last departure that will get me to London Liverpool Street before sunset?

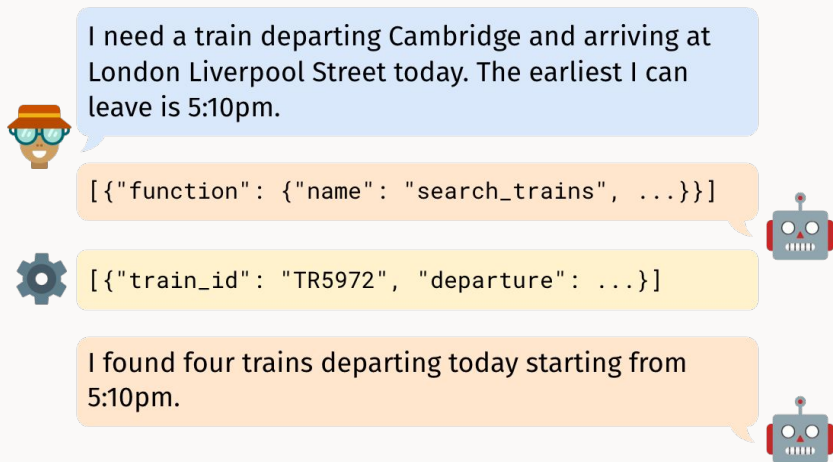
What is task-oriented dialogue?

TOD systems help users with specific tasks through natural language interaction, typically querying databases or calling APIs



What is task-oriented dialogue?

LLMs in end-to-end TOD systems generate both **natural language responses** and **structured tool calls**



Why task-oriented dialogue?

(1) Inherently multi-turn



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I found four trains departing today starting from 5:10pm.



What time is the last departure that will get me to London Liverpool Street before sunset?

The last train that will get you to London Liverpool Street before sunset (9:20pm) departs at 7:59pm and arrives at 9:27pm. Would you like to proceed with booking tickets for this train?

Why task-oriented dialogue?

- (1) Inherently multi-turn
- (2) Naturally allows testing various capabilities
 - commonsense



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- (1) Inherently multi-turn
- (2) Naturally allows testing various capabilities
 - commonsense
 - spatial and temporal reasoning



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 - spatial and temporal reasoning
 - reasoning over structured data



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 - commonsense
 - spatial and temporal reasoning
 - reasoning over structured data
 - arithmetic, etc.



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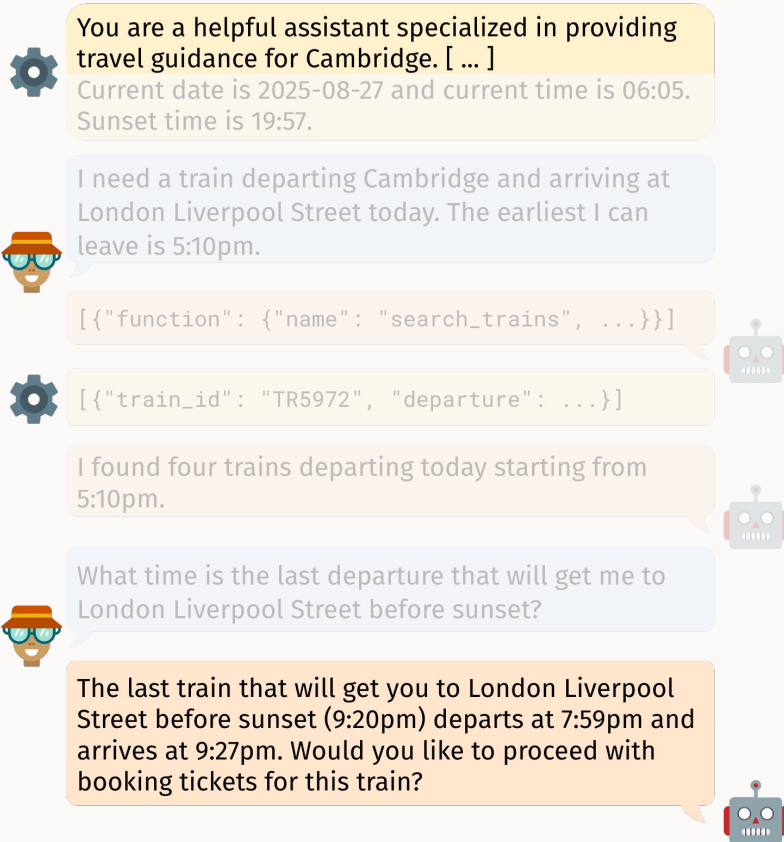
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Why task-oriented dialogue?

- (1) Inherently multi-turn
- (2) Naturally allows testing various capabilities
 - commonsense
 - spatial and temporal reasoning
 - reasoning over structured data
 - arithmetic, etc.
- (3) Generation constrained by format and/or style



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The BOULDER benchmark

The benchmark has the following properties:



Dynamic, to mitigate data contamination



Automatically verifiable answers



Controlled comparison of standalone and TOD settings

Tasks

Task	Domains	Example
Ticket price	Trains	Can you first calculate the total price for me? There are four of us. Three of us are buying return tickets, one standard class, and two first class - one with a 33% discount, the other without discount. Another person is buying a one-way ticket.
Booking price	Hotels	Can you first calculate the total price for me? It's for five people staying for four nights. One person will stay in a single room and the others will share double rooms. One person in a double room will check in one day later.
Departure time	Trains	What time is the last departure that will get me to Cambridge before sunset?
Departure freq.	Trains	How often on average do trains run between 19:00 and midnight?
Opening hours	Restaurants	Which of them are open on Wednesday for the entire time between 2pm and 6:30pm?
Distance	Hotels, Restaurants	How far is it from the hotel to the restaurant?
Directional relations	Attractions, Restaurants	Is Museum of Archaeology and Anthropology south of Pizza Express?
Shortest path	Attractions, Hotels	I want to visit Broughton House Gallery, King's College, and Camboats. I'll walk between them starting from the hotel and taking a taxi back from the last one. What order should I visit all of them in to minimize my walking distance?

Tasks

arithmetic

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Tasks

temporal	Task	Domains	Example
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Tasks

formal and commonsense aspects

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How do we generate test examples?

BOULDER consists of three main components:



Adapted and extended
database from MultiWOZ
(Budzianowski et al. 2018)

+



Dialogue templates with
placeholders and multiple
paraphrases

+



Python code to
automatically generate the
benchmark

MultiWOZ database

Source of in-domain data, adapted and extended for our benchmark

MultiWOZ - A Large-Scale Multi-Domain Wizard-of-Oz Dataset for Task-Oriented Dialogue Modelling

Pawel Budzianowski¹, Tsung-Hsien Wen^{2*}, Bo-Hsiang Tseng¹,
Iñigo Casanueva¹, Stefan Ultes¹, Osman Ramadan¹ and Milica Gasić¹
¹Department of Engineering, University of Cambridge, UK, ²PolyAI, UK
{pfb30,mg436}@cam.ac.uk

Abstract

Even though machine learning has become the major scene in dialogue research community, the real breakthrough has been blocked by the scale of data available. To address this fundamental obstacle, we introduce the Multi-Domain Wizard-of-Oz dataset (MultiWOZ), a fully-labelled collection of human-human written conversations spanning over multiple domains and topics. As a size of UK dialogues, it is at least one order of magnitude larger than all previous annotated task-oriented corpora. The contribution of this work apart from the open-sourced dataset labelled with dialogue belief states and dialogue actions is two-fold: firstly, a detailed description of the data collection procedure along with a summary of data structure and analysis is provided. The proposed data-collection pipeline is entirely based on crowd-sourcing without the need of hiring professional annotators; secondly, a set of benchmark results of belief tracking, dialogue act and response generation is reported, which shows the usability of the data and sets a baseline for future studies.

1 Introduction

Conversational Artificial Intelligence (Conversational AI) is one of the long-standing challenges in computer science and artificial intelligence since the Dartmouth Proposal (McCarthy et al., 1955). As human conversation is inherently complex and ambiguous, learning an open-domain conversational AI that can carry on arbitrary tasks is still very far-off (Vinyals and Le, 2015). As a consequence, instead of focusing on creating ambitious conversational agents that can reach human-level intelligence, industrial practice has focused on building task-oriented dialogue systems (Young et al., 2013) that can help with specific tasks such as flight reservation (Seneff and Polifroni, 2000)

*The work was done while at the University of Cambridge.

or bus information (Raux et al., 2005). As the need of hands-free use cases continues to grow, building a conversational agent that can handle tasks across different application domains has become more and more prominent (Ram et al., 2018).

Dialogues systems are inherently hard to build because there are several layers of complexity: the noise and uncertainty in speech recognition (Black et al., 2011); the ambiguity when understanding human language (Williams et al., 2013); the need to integrate third-party services and dialogue context in the decision-making (Traum and Larson, 2003; Paek and Pieraccini, 2008); and finally, the ability to generate natural and engaging responses (Stent et al., 2005). These difficulties have led to the same solution of using statistical framework and machine learning for various system components, such as natural language understanding (Henderson et al., 2013; Mesnil et al., 2015; Mikić et al., 2017a), dialogue management (Galić and Young, 2014; Tegho et al., 2018), language generation (Wen et al., 2015; Kiddon et al., 2016), and even end-to-end dialogue modelling (Zhao and Eskenazi, 2016; Wen et al., 2017; Eric et al., 2017).

To drive the progress of building dialogue systems using data-driven approaches, a number of conversational corpora have been released in the past. Based on whether a structured annotation scheme is used to label the semantics, these corpora can be roughly divided into two categories: corpora with structured semantic labels (Hemphill et al., 1990; Williams et al., 2013; Asti et al., 2017; Wen et al., 2017; Eric et al., 2017; Shah et al., 2018); and corpora without semantic labels but with an implicit user goal in mind (Ritter et al., 2010; Lowe et al., 2015). Despite these efforts, aforementioned datasets are usually constrained in one or more dimensions such as missing proper annotations, only available in a limited capacity, lacking multi-domain use cases, or having a negli-

Domain	Categorical slots	Non-categorical slots	Intents
Restaurant	pricerange, area, bookday, bookpeople	food, name, booktime, address, phone, postcode, ref	find, book
Attraction	area, type	name, address, entrancefee, openhours, entrancefee, openhours, phone, postcode	find
Hotel	pricerange, parking, internet, stars, area, type, bookpeople, bookday, bookstay	name, address, phone, postcode, ref	find, book
Taxi	-	destination, departure, arriveby, leaveat, phone, type	book
Train	destination, departure, day, bookpeople	arriveby, leaveat, trainid, ref, price, duration	find, book
Bus	day	departure, destination, leaveat	find
Hospital	-	department, address, phone, postcode	find
Police	-	name, address, phone, postcode	find

MultiWOZ database

Source of in-domain data, adapted and extended for our benchmark

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→ Hotel	pricerange, parking, internet, stars, area, type, bookpeople, bookday, bookstay	name, address, phone, postcode, ref	find, book
Taxi	-	destination, departure, arriveby, leaveat, phone, type	book
→ Train	destination, departure, day, bookpeople	arriveby, leaveat, trainid, ref, price, duration	find, book
Bus	day	departure, destination, leaveat	find
Hospital	-	department, address, phone, postcode	find
Police	-	name, address, phone, postcode	find

Dialogue templates

Human-written conversation history for each task + LLM paraphrases

```
[
  {
    "role": "assistant",
    "content": "Hi, how can I help you today?"
  },
  {
    "role": "user",
    "content": "I'm looking for {{ food }} restaurants in the {{ area_adj }} part of town."
  },
  {
    "role": "placeholder",
    "content": "tool_call_restaurant"
  },
  {
    "role": "placeholder",
    "content": "tool_response_restaurant"
  },
  {
    "role": "assistant",
    "content": "I found {{ num_restaurants }} {{ food }} restaurants in the {{ area }}. Do you have any other preferences to help narrow this down?"
  },
  {
    "role": "user",
    "content": "Which of them are open on {{ weekday }} {{ time_specification }}?"
  }
]
```

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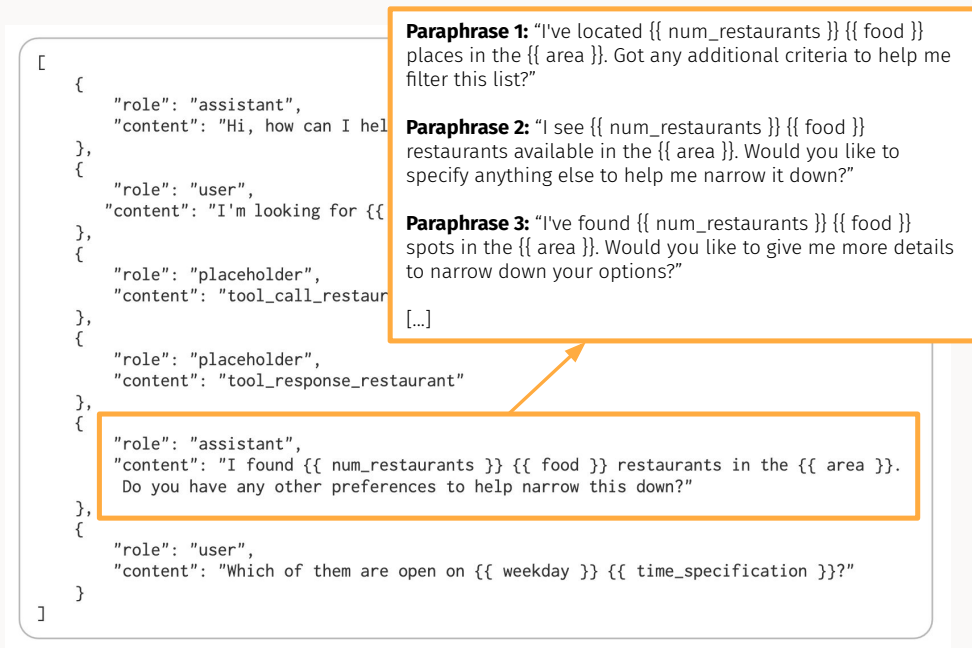
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]
```

Paraphrase 1: "I've located {{ num_restaurants }} {{ food }} places in the {{ area }}. Got any additional criteria to help me filter this list?"

Paraphrase 2: "I see {{ num_restaurants }} {{ food }} restaurants available in the {{ area }}. Would you like to specify anything else to help me narrow it down?"

Paraphrase 3: "I've found {{ num_restaurants }} {{ food }} spots in the {{ area }}. Would you like to give me more details to narrow down your options?"

[...]



Test item generators

Procedures in Python used to generate new items from DB and templates

```
generator = TrainDepartureTimeGenerator(rng, database)
```

```
x, y = generator(dialogue_templates, n=100)
```

Experimental setup

- (1) **Baseline:** problem presented in isolation
- (2) **Dialogue:** embedded in TOD (implementation adapted from Xu et al. (2024))
- (3) **Dialogue-concise:** *Dialogue* + length constraint

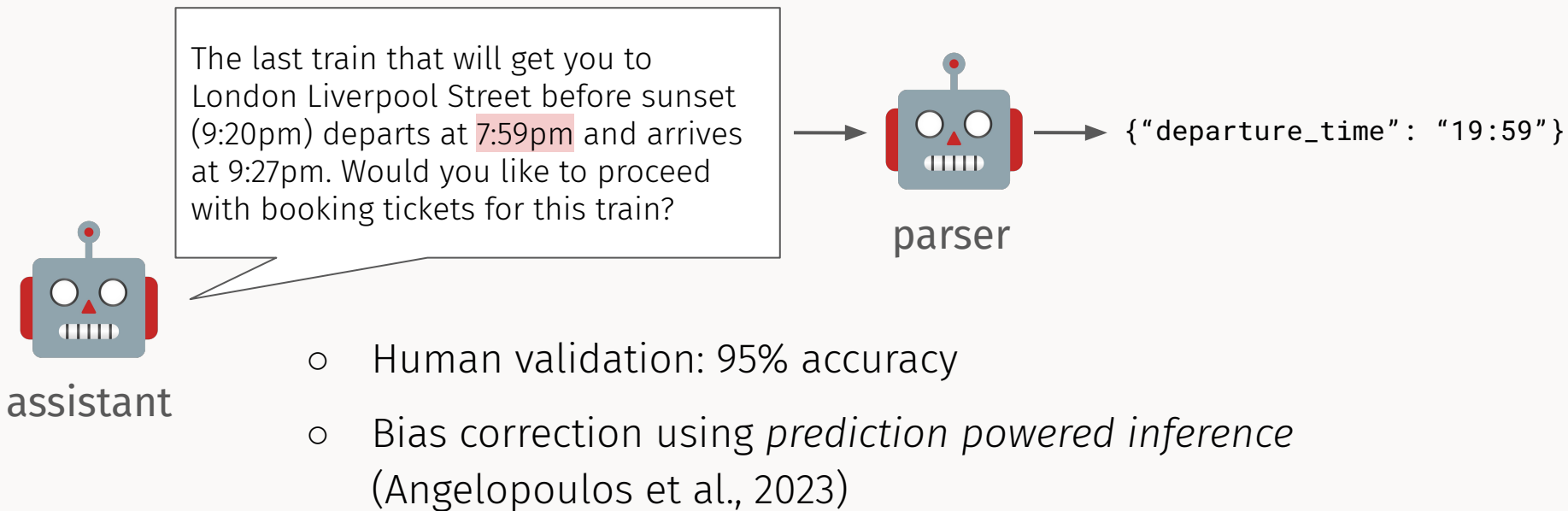
Evaluated models:

- 6 open-weight LLMs with 4B to 671B parameters (Qwen3, Mistral Small, etc.)
- 2 proprietary LLMs (Gemini 2.5 Flash, Claude Sonnet 4.5)

Metrics depending on tasks: accuracy, precision, mean absolute error

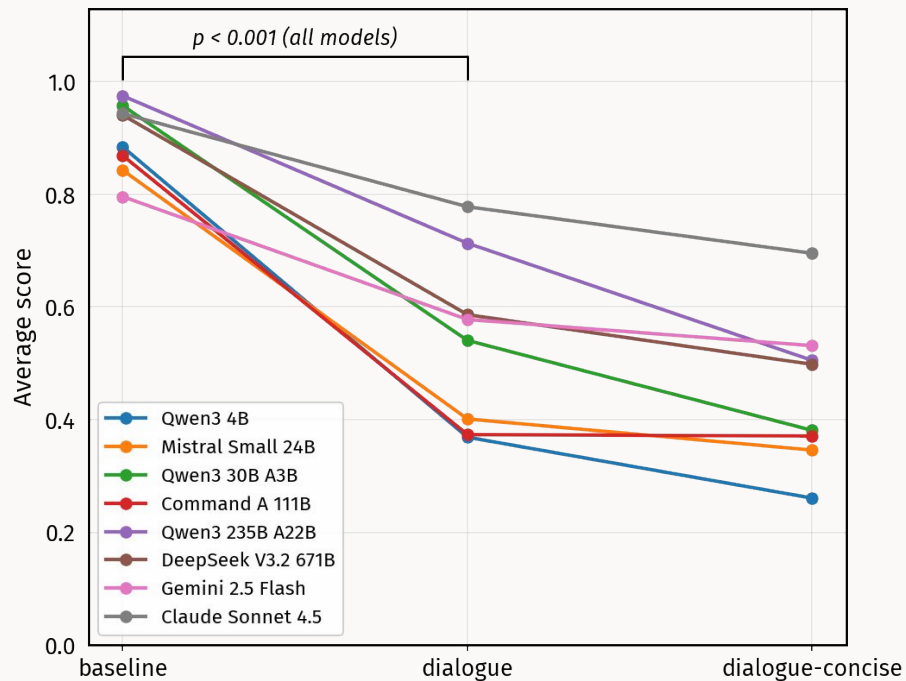
Answer extraction

- Specialized LLM-based parser for each semantic category

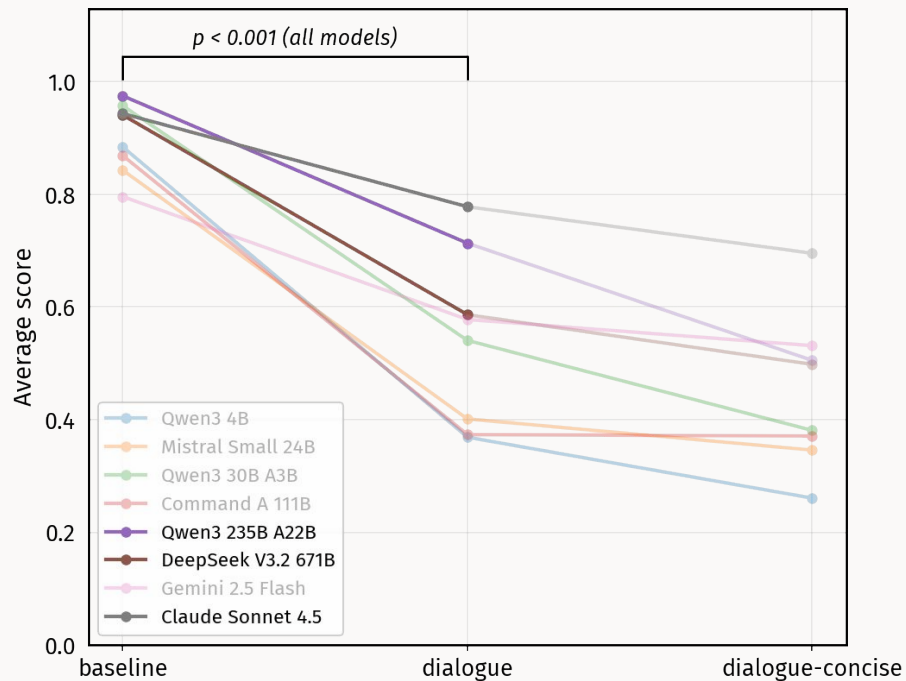


Results

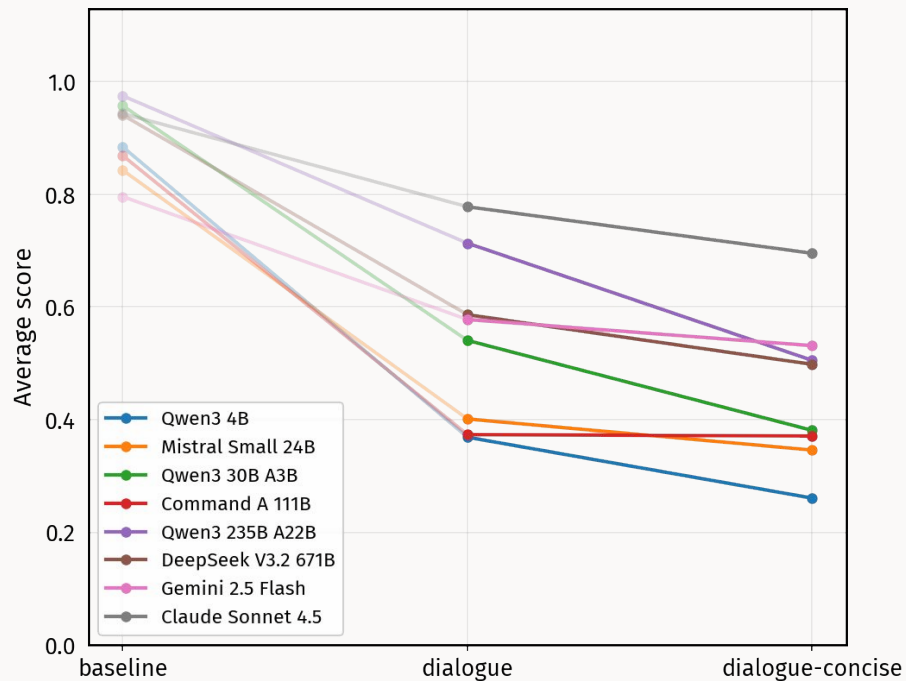
Substantial gap between the *baseline* and *dialogue* setups



Remains significant even with large and closed LLMs



Average *dialogue* scores relatively close to the *concise* setup



Ablation experiments

Does multi-turn interaction make results worse?

Ablation 1: *dialogue* setup transformed into a *single-turn* variant

SYSTEM: You are a helpful assistant specialized in providing travel guidance for Cambridge. Your task is to help users find venues or transport based on their preferences, and make bookings and reservations. Based on the user's requirements, there are four different sub-tasks: restaurants, hotels, attractions, and trains.

You can use the available tools to search for venues or transport, and make reservations or purchases. Based on the user's input, either respond with a message or call a tool.

[...]

ASSISTANT: Hi, how can I help you today?

USER: I'm looking for trains from Broxbourne to Cambridge on Thursday.

ASSISTANT: `{"type": "function", "id": "205d4e", "function": {"name": "search_trains", "arguments": {"departure": "broxbourne", "destination": "cambridge", "weekday": "thursday"}}`

TOOL: `[{"trainID": "TR0393", "departure": "broxbourne", "destination": "cambridge", "day": "thursday", "leaveAt": "06:07", "arriveBy": "07:07", "duration": "60 minutes", "price_standard": "18.40 pounds", "price_first_class": "27.60 pounds"}, ...]`

ASSISTANT: I found 59 trains leaving on Thursday. Do you have a preferred departure time?

USER: How often on average do trains run between 20:00 and midnight?

Does multi-turn interaction make results worse?

Ablation 1: *dialogue* setup transformed into a *single-turn* variant

SYSTEM: You are a helpful assistant specialized in providing travel guidance for Cambridge. Your task is to help users find venues or transport based on their preferences, and make bookings and reservations. Based on the user's requirements, there are four different sub-tasks: restaurants, hotels, attractions, and trains.

You can use the available tools to search for venues or transport, and make reservations or purchases. ~~Based on the user's input, either respond with a message or call a tool.~~ Based on the user's input, either answer their questions or confirm a booking, purchase or reservation. We will parse your confirmation and make the actual booking, purchase or reservation later.

[...]

Data

```
[{"trainID": "TR0393", "departure": "broxbourne", "destination": "cambridge", "day": "thursday",  
"leaveAt": "06:07", "arriveBy": "07:07", "duration": "60 minutes", "price_standard": "18.40 pounds",  
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ASSISTANT: Hi, how can I help you today?

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ASSISTANT: I found 59 trains leaving on Thursday. Do you have a preferred departure time?

USER: ~~How often on average do trains run between 20:00 and midnight?~~

USER: I'm looking for trains from Broxbourne to Cambridge on Thursday. How often on average do they run between 20:00 and midnight?

Does multi-turn interaction make results worse?

Ablation 2: *baseline* setup transformed into a *multi-turn* variant

USER: Given the list of trains in JSON format below, calculate how often on average do trains run between 20:00 and midnight.

Trains:

```
[{"trainID": "TR0393", "departure": "broxbourne", "destination": "cambridge", "day": "thursday",  
"leaveAt": "06:07", "arriveBy": "07:07", "duration": "60 minutes", "price_standard": "18.40 pounds",  
"price_first_class": "27.60 pounds"}, ...]
```

Does multi-turn interaction make results worse?

Ablation 2: *baseline* setup transformed into a *multi-turn* variant

USER:SYSTEM: Given the list of trains in JSON format below, ~~calculate how often on average do trains run between 20:00 and midnight.~~ answer the user's questions.

Trains:

```
[{"trainID": "TR0393", "departure": "broxbourne", "destination": "cambridge", "day": "thursday",  
"leaveAt": "06:07", "arriveBy": "07:07", "duration": "60 minutes", "price_standard": "18.40 pounds",  
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ASSISTANT: Hi, how can I help you today?

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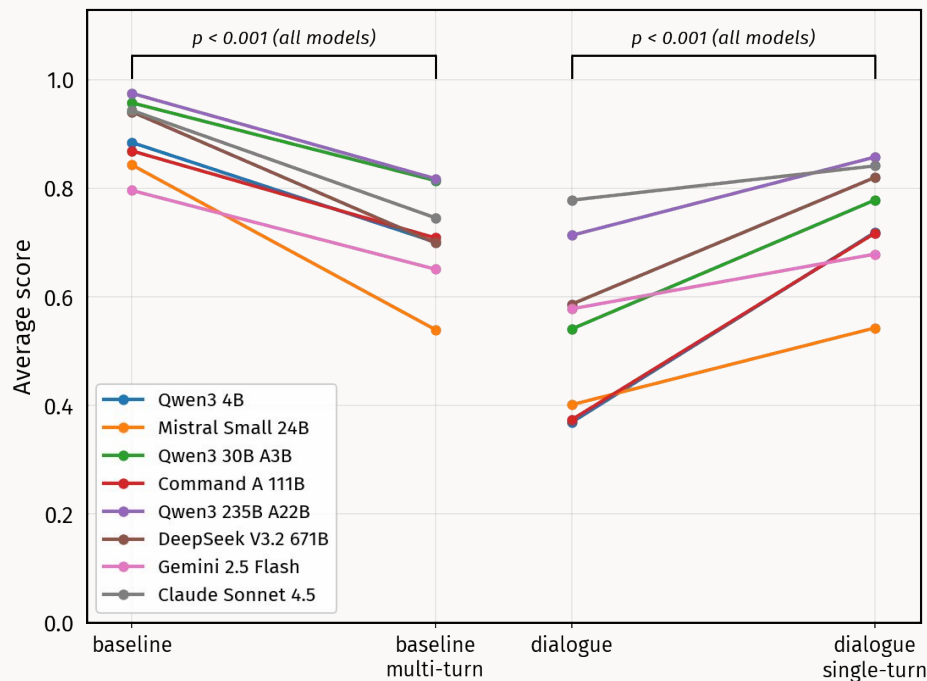
USER: How often on average do trains run between 20:00 and midnight?

Does multi-turn interaction make results worse?

→ Multi-turn interaction harms performance for all models in both setups

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How does assistant persona affect performance?

Ablation 1: *baseline* setup with TOD role instructions

USER: Given the list of trains in JSON format below, calculate how often on average do trains run between 20:00 and midnight.

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[{"trainID": "TR0393", "departure": "broxbourne", "destination": "cambridge", "day": "thursday", "leaveAt": "06:07", "arriveBy": "07:07", "duration": "60 minutes", "price_standard": "18.40 pounds", "price_first_class": "27.60 pounds"}, ...]

How does assistant persona affect performance?

Ablation 1: *baseline* setup with TOD role instructions

USER: You are a helpful assistant specialized in providing travel guidance for Cambridge. Always maintain a friendly, polite, and helpful tone throughout the conversation. Given the list of trains in JSON format below, calculate how often on average do trains run between 20:00 and midnight.

Trains:
[{"trainID": "TR0393", "departure": "broxbourne", "destination": "cambridge", "day": "thursday", "leaveAt": "06:07", "arriveBy": "07:07", "duration": "60 minutes", "price_standard": "18.40 pounds", "price_first_class": "27.60 pounds"}, ...]

How does assistant persona affect performance?

Ablation 2: *dialogue* setup with instructions to use more tokens for reasoning

SYSTEM: You are a helpful assistant specialized in providing travel guidance for Cambridge. Your task is to help users find venues or transport based on their preferences, and make bookings and reservations. Based on the user's requirements, there are four different sub-tasks: restaurants, hotels, attractions, and trains.

You can use the available tools to search for venues or transport, and make reservations or purchases. Based on the user's input, either respond with a message or call a tool.

Start by greeting the user and asking what task they want to perform: "Hi, how can I help you today?"

[...]

How does assistant persona affect performance?

Ablation 2: *dialogue* setup with instructions to use more tokens for reasoning

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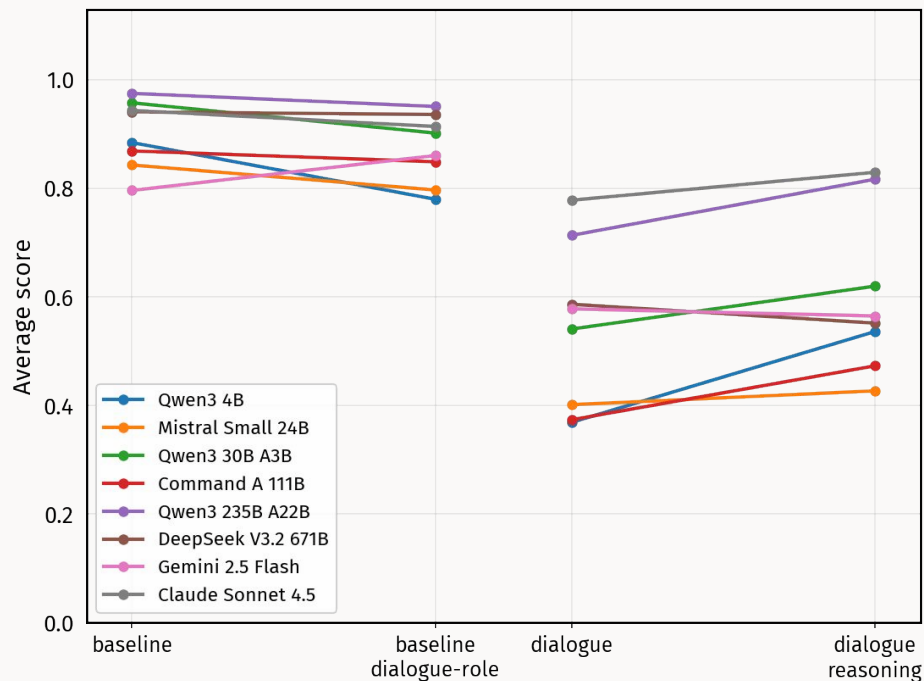
You can use the available tools to search for venues or transport, and make reservations or purchases. Based on the user's input, either respond with a message or call a tool. When the answer requires reasoning, take more time to think about the answer.

Start by greeting the user and asking what task they want to perform: "Hi, how can I help you today?"

[...]

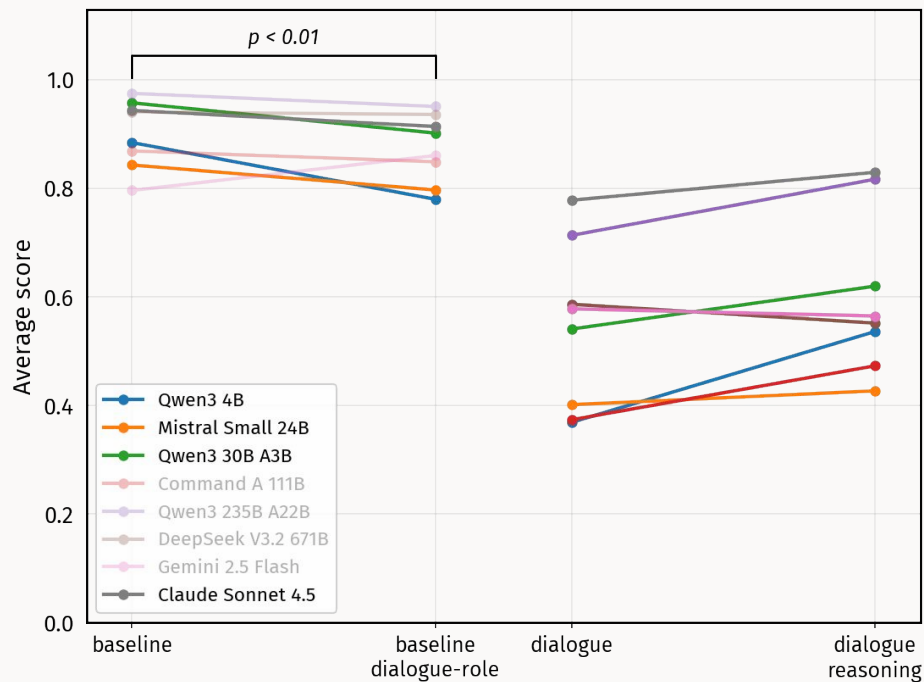
How does assistant persona affect performance?

Role instructions have an effect, although not uniform



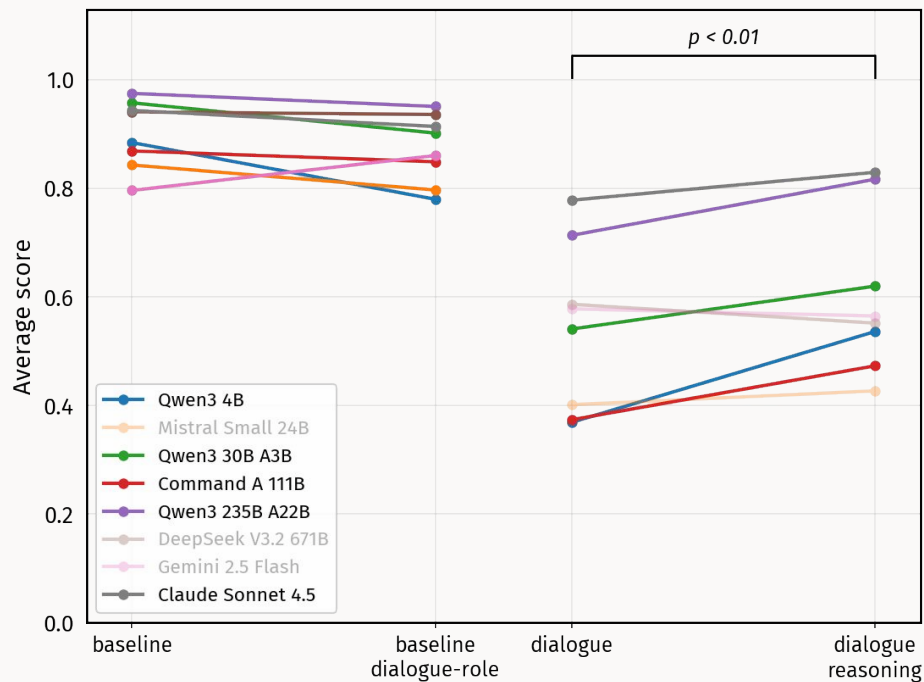
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Role instructions have an effect, although not uniform



How does assistant persona affect performance?

Asking to use more tokens helps, but does not recover baseline performance



What if we use a subset of domains and no tools?

Ablation 1: dialogue setup with only the domains needed for the particular task

You are a helpful assistant specialized in providing travel guidance for Cambridge. Your task is to help users find venues or transport based on their preferences, and make bookings and reservations. Based on the user's requirements, there are four different sub-tasks: restaurants, hotels, attractions, and trains.

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Start by greeting the user and asking what task they want to perform: "Hi, how can I help you today?"

Always maintain a friendly, polite, and helpful tone throughout the conversation.

Today is {{ weekday }} {{ date }} and the current time is {{ time }}.

Task 1: Restaurant

[...]

Task 2: Hotel

[...]

Task 3: Attraction

[...]

Task 4: Train

[...]

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What if we use a subset of domains and no tools?

Ablation 2: dialogue setup without tools

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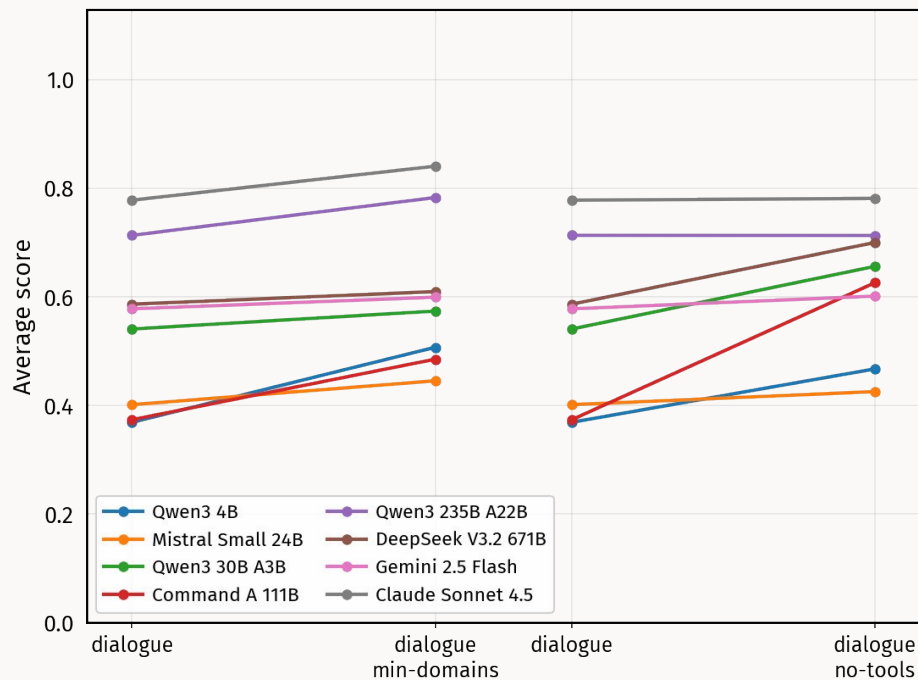
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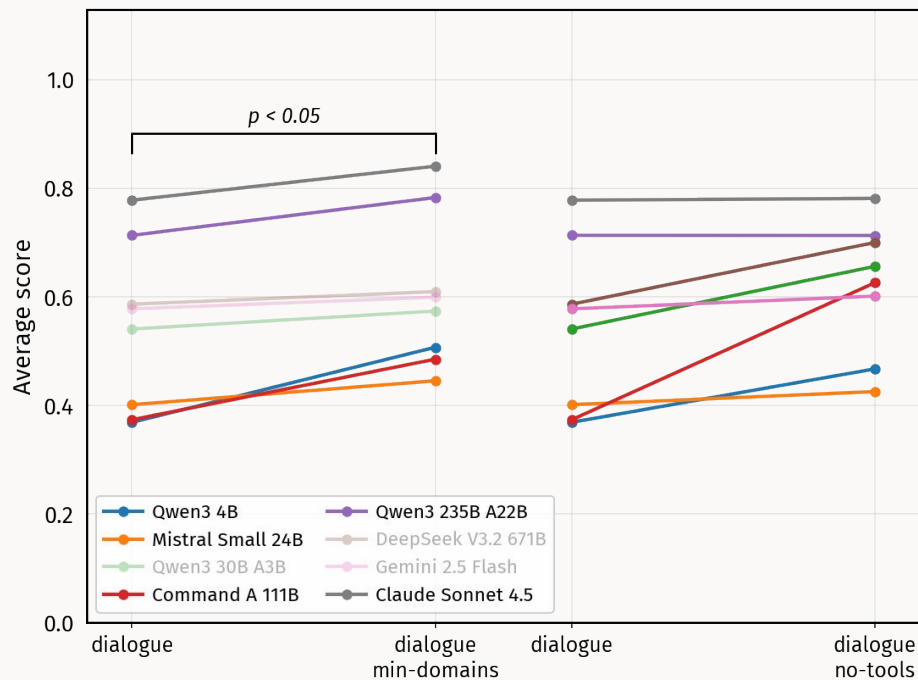
What if we use a subset of domains and no tools?

Reducing domains in context helps slightly for some models



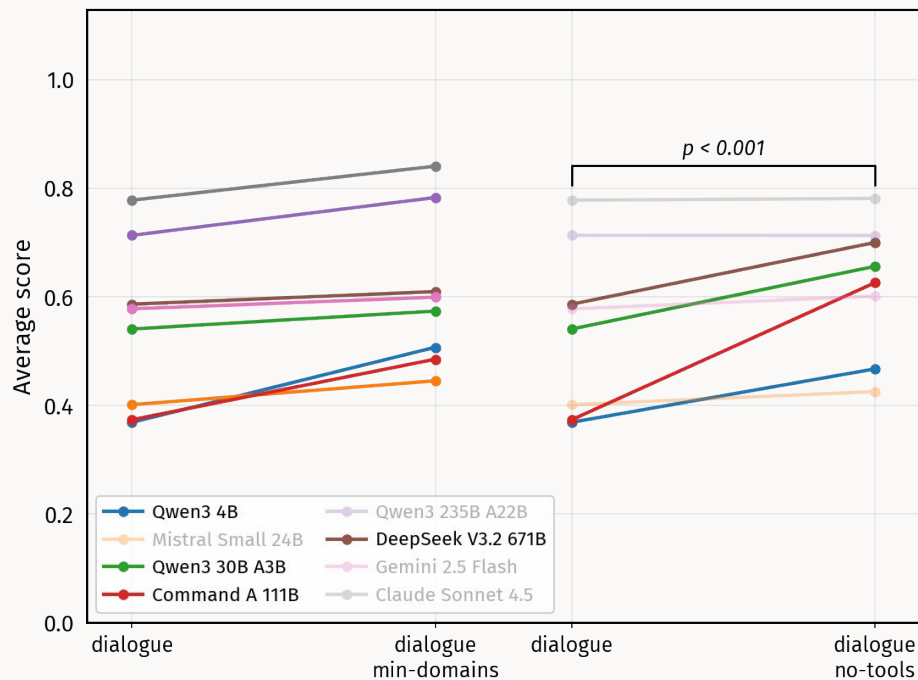
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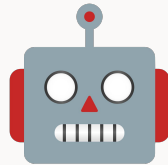
Similarly, some models do better without tool schemas and messages



Qualitative analysis

Examples of failure modes

Quick answer with post-hoc explanation



The total cost of your stay at Hamilton Lodge is £1,024.

Here's how I calculated this:

- 5 nights x 3 single rooms = £750

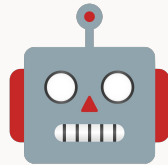
- 5 nights x 1 double room = £365

- 1 night x 1 double room = £73

Total cost = £750 + £365 + £73 = £1,024

Examples of failure modes

Quick answer with post-hoc explanation



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Examples of failure modes

Heuristics and templates



Trains leave Broxbourne for Cambridge
at 19:27, 20:57, 22:27 and 23:57.
That's roughly every two hours.

Examples of failure modes

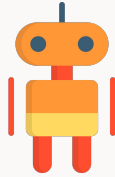
Heuristics and templates



Trains leave Broxbourne for Cambridge
at 19:27, 20:57, 22:27 and 23:57.
That's roughly every two hours.

Examples of failure modes

Refusals



I am sorry, I cannot fulfill this request.
I do not have the ability to calculate
the distance between the hotel and
the restaurant.

Examples of failure modes

Rule inversion

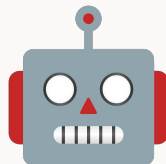
Let me help you calculate the total price for two passengers:

- First passenger (first class, one-way with 60% discount):

Original first class price: £28.70

With 60% discount: $£28.70 \times 0.6 = £17.22$
(after applying 60% discount)

[...]



Examples of failure modes

Rule inversion

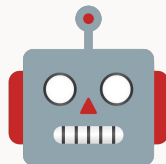
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(after applying 60% discount)

[...]



Examples of failure modes

Post-hoc explanation + fabricated data

Between 19:00 and 23:30 on Friday, trains run every 30 minutes. Here are the departures during that time:

- 19:16
- 19:46
- 20:16
- 20:46
- 21:16
- 21:46
- 22:16
- 22:46
- 23:16

So, on average, a train departs every 30 minutes. Let me know if you'd like to book a ticket or need further details!



Examples of failure modes

Post-hoc explanation + fabricated data

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- 19:16

- 19:46

- 20:16

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- 21:16

- 21:46

- 22:16

- 22:46

- 23:16

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Summary

- LLMs show a substantial gap in reasoning between standalone and TOD setups
- Partially explained by multi-turn interaction, but also persona bias and tool use
- Integrating prompted LLMs in TOD systems with reasoning might not be straightforward
- LLMs need to be evaluated more in realistic and interactive scenarios

**Followup: Reasoning models
+ instruction following**

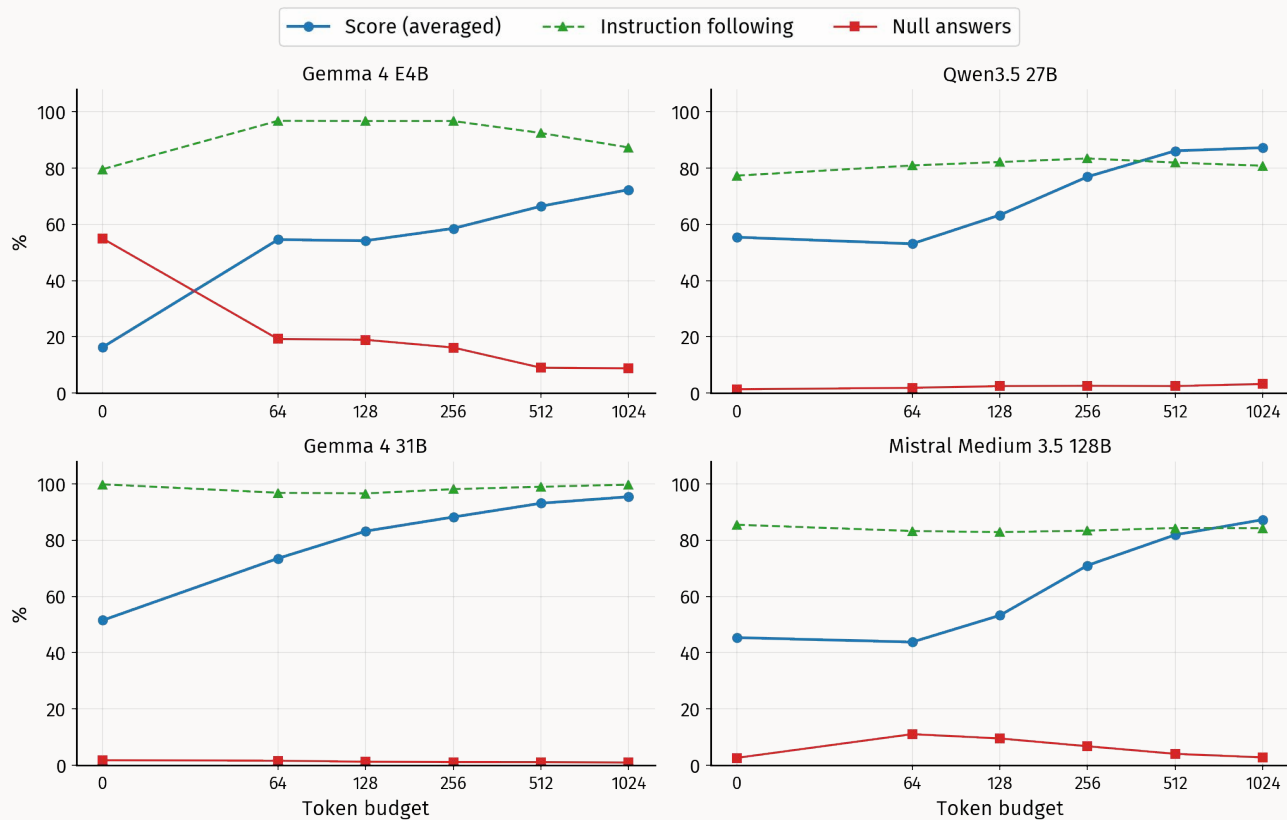
Reasoning models should solve this, right?

- Probably yes, but they can be inefficient
 - Thousands of output tokens
 - Latency (especially for spoken dialogue)
- Can we improve this with budget forcing? (Muennighoff et al., 2025)
 - Set token budget, force final answer if exceeded

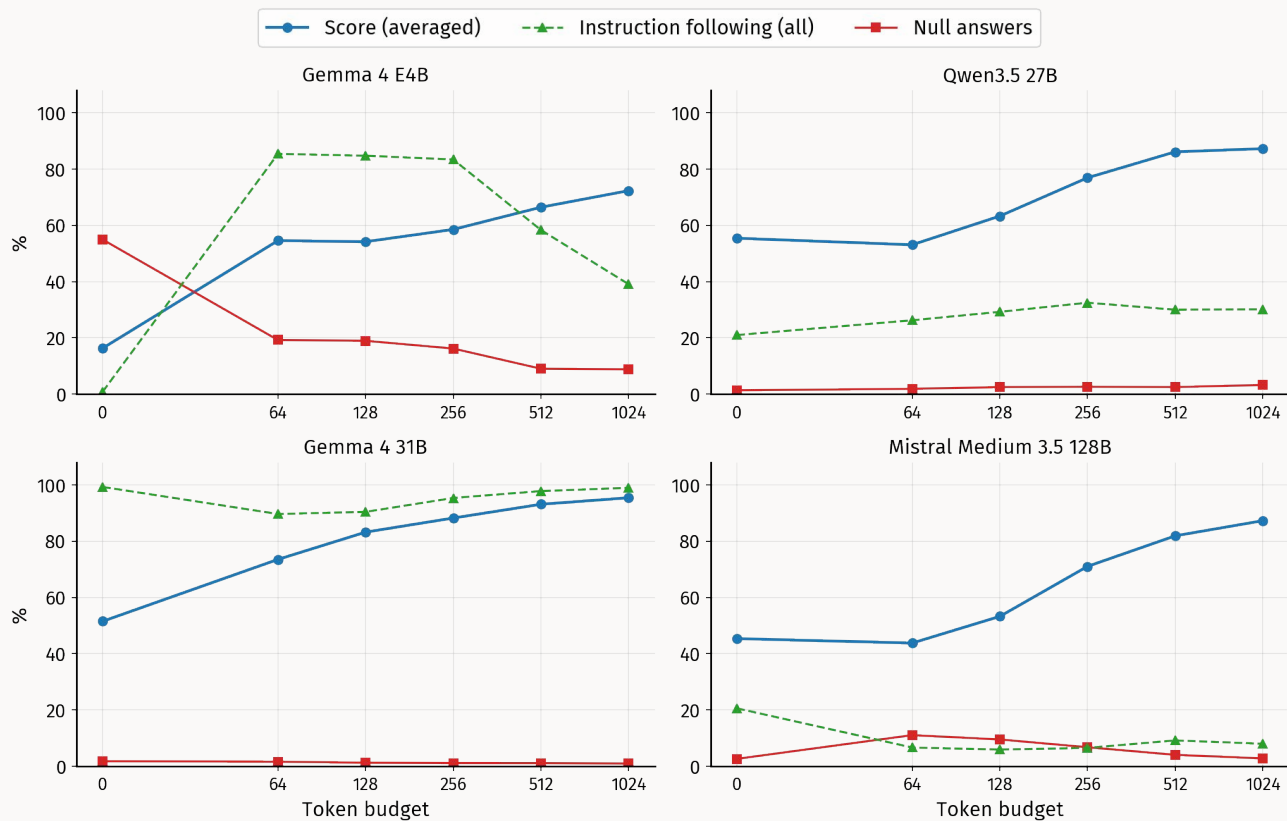
Reasoning in TOD with token budget

- Token budgets: 0, 64, 128, 256, ...
- If budget hit, we append a prefill + `</think>` token:
 - `“Given the time constraints, I should stop reasoning here and provide my answer based on what I have so far.</think>”`
- Measure reasoning accuracy and refusal rate for each budget
- Additional constraints:
 - Instruction following (length, keywords, formatting, etc.)
 - Multilingual (English, Czech, Spanish, Chinese)

Preliminary results (English)



Preliminary results (English)



TOD with reasoning models: summary

- Substantial amount of tokens may be needed for good performance
 - Can be too inefficient, depending on setup (e.g. spoken dialogue systems)
- Even larger budgets may not be enough for reasoning + instruction following
- But this is very much a work in progress, more experiments are needed

Thank you!

Contact

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<https://ivankartac.github.io>

Links

Paper: <https://aclanthology.org/2026.acl-long.560>

Repo: <https://github.com/ivankartac/boulder>

Slides: <https://bit.ly/nasknlp-ik>

Thanks

Mateusz Lango

Ondřej Dušek

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Heng-Da Xu, Xian-Ling Mao, Puhai Yang, Fanshu Sun, and Heyan Huang. 2024. [Rethinking Task-oriented Dialogue Systems: From Complex Modularity to Zero-shot Autonomous Agent](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2748–2763, Bangkok, Thailand.